Sahaj Bhadja

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Summary

Full-stack engineer building **accessible React applications, Python APIs, SQL databases**, and automated tests.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Data Structures, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Web Technologies, Computer Networks, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built an Azure-backed education platform with normalized schemas, row-level isolation, validation controls, and indexed **SQL queries**, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users.
- Reworked asynchronous Python services with FastAPI workers, explicit job states, error handling, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **application reliability** daily.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed responsive **React and Flask features** for uploads, grading history, rubrics, and feedback, reliably supporting 150+ instructors and 10K+ daily requests through production REST APIs securely at scale.
- Designed concurrent file-processing services with secure PDF, DOCX, and ZIP ingestion, automated tests, provider fallbacks, and CI/CD deployments, reducing feedback turnaround 60% during peak **browser application** workloads.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed cross-platform integration modules using object-oriented patterns and REST APIs, improving response times 30%, reducing technical debt 20%, and strengthening **maintainable business logic** across distributed application services.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code reviews**, defect resolution, Git collaboration, and dependable releases across Windows, Linux, and macOS.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight MERN microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records across distributed locations through **responsive interfaces, REST APIs**, WebSockets, and robust data validation.
- Developed React and TypeScript workflows for real-time records and third-party integrations, while Jenkins CI/CD, Docker staging, and **automated testing** reduced deployment time 50% across controlled production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built React and TypeScript booking workflows for scheduling, reservation tracking, coupon pricing, and payment status, delivering **accessible browser interfaces** backed by sub-200-millisecond APIs across 50K+ service records.
- Designed data models and integration contracts for 12+ endpoints, aligning business requirements with analysts through wireframes and technical documentation while reducing **application design errors** 25% before implementation.

Skills

Languages: JavaScript, TypeScript, Python, Java, SQL, C#, C++, HTML5, CSS3, Bash
Frontend: React, Next.js, Angular, Responsive Design, Accessibility, AJAX, Tailwind CSS, Wireframes, Browser APIs
Backend & Data: Node.js, Express.js, Flask, FastAPI, REST APIs, Microservices, PostgreSQL, SQL Server, MySQL, MongoDB, SQLite, Data Modeling, Query Optimization, Authentication, Authorization
Engineering: Git, GitHub, Docker, Azure, AWS, CI/CD, Jenkins, GitHub Actions, Unit Testing, Integration Testing, PyTest, Mockito, Jest, Code Review, Pair Programming, Debugging, Error Handling, Data Validation, Security Controls, Agile, SDLC, Technical Documentation, AI-Assisted Development

Projects

DonorFlow Portal | *React, TypeScript, Node.js, PostgreSQL, Jest, WCAG*

- Designed a responsive, accessible donor workflow with reusable React components, secure Node.js APIs, relational schemas, validation, error handling, role-based access controls, and **automated unit and integration tests**.

Hospital Inventory Exchange | *React, Python, FastAPI, PostgreSQL, REST APIs, Docker*

- Architected a browser-based ordering and inventory application connecting hospital workflows with internal services through **REST integrations**, transactional updates, audit trails, responsive interfaces, documented APIs, and deployment-ready containers.

Hospital Management Information System | *React, Node.js, MongoDB, ABDM APIs, Sentry, Twilio*

- Built a healthcare management application with responsive React workflows, secure REST APIs, patient and appointment records, ABDM interoperability, Twilio notifications, and **Sentry-based monitoring** for reliable issue diagnosis.